

Distress signal

A **distress signal**, also known as a **distress call**, is an internationally recognized means for obtaining help. Distress signals are communicated by transmitting radio signals, displaying a visually observable item or illumination, or making a sound audible from a distance.

A distress signal indicates that a person or group of people, watercraft, aircraft, or other vehicle is threatened by a serious or imminent danger and requires immediate assistance.^{[1]:PCG D-3} Use of distress signals in other circumstances may be against local or international law. An urgency signal is available to request assistance in less critical situations.

For distress signalling to be the most effective, two parameters must be communicated:

- Alert or notification of an emergency in progress
- Position or location (or localization or pinpointing) of the party in distress.

For example, a single aerial flare alerts observers to the existence of a vessel in distress somewhere in the general direction of the flare sighting on the horizon but extinguishes within one minute or less. A hand-held flare burns for three minutes and can be used to localize or pinpoint more precisely the exact location or position of the party in trouble. An EPIRB both notifies or alerts authorities and at the same time provides position indication information.

Maritime

Distress signals at sea are defined in the International Regulations for Preventing Collisions at Sea and in the International Code of Signals.^[2] Mayday signals must only be used where there is grave and imminent danger to life. Otherwise, urgent signals such as pan-pan can be sent. Most jurisdictions have large penalties for false, unwarranted, or prank distress signals. The alerts are of utmost importance in ensuring the safety of life at sea, and are governed by international maritime law, specifically the International Convention for the Safety of Life at Sea (SOLAS).^[3]

Distress can be indicated by any of the following officially sanctioned methods:

- Transmitting a spoken voice Mayday message by radio over very high frequency channel 16 (156.8 MHz) or medium frequency on 2182 kHz^[2]
- Transmitting a digital distress signal by activating (or pressing) the distress button on a marine radio equipped with Digital Selective Calling (DSC) over VHF channel 70 or over another designated DSC frequency in the maritime MF and HF bands.^[2]
- Transmitting a digital distress signal by activating (or pressing) the distress button (or key) on an Inmarsat-C satellite internet device^[2]
- Sending the Morse code group SOS (···—····) by light flashes or sounds^[2]
- Burning a red flare (either hand-held or aerial parachute flare)^[2]
- Launching distress rockets^[2]



Distress signals

- Emitting orange smoke from a canister^[2]
- Showing flames on the vessel (as from a burning tar barrel, oil barrel, etc.)^[2]
- Raising and lowering slowly and repeatedly both arms outstretched to each side^[2]
- Making a continuous sound with any fog-signaling apparatus^[2]
- Firing a gun or other explosive signal at intervals of about a minute^[2]
- Flying the international maritime signal flags **NC** ^[2]
- Displaying a visual signal consisting of a square flag having above or below it a ball or anything resembling a ball (round or circular in appearance)^[2]

A floating man-overboard pole or dan buoy can be used to indicate that a person is in distress in the water and is ordinarily equipped with a yellow and red flag (international code of signals flag "O") and a flashing lamp or strobe light.



Smoke signal

In North America, marine search and rescue agencies in Canada and the United States also recognize certain other distress signals:

- Sea marker dye
- White high-intensity strobe light flashing at 60 times per minute

Automated radio signals

In addition, distress can be signaled using automated radio signals such as a Search and Rescue Transponder (SART) which response to 9 GHz radar signal, or an Emergency Position-Indicating Radio Beacon (EPIRB) which operates in the 406 MHz radio frequency. EPIRB signals are received and processed by a constellation of satellites known as Cospas-Sarsat. Older EPIRBs that use 121.5 MHz are obsolete. Many regulators require vessels that proceed offshore to carry an EPIRB.

Many EPIRBs have an in-built Global Positioning System receiver. When activated these EPIRBs rapidly report the latitude and longitude of the emergency accurate to within 120 m (390 ft). The position of non-GPS EPIRBs is determined by the orbiting satellites, this can take ninety minutes to five hours after activation and is accurate to within 5 km (3.1 mi). Marine safety authorities recommend the use of GPS-equipped EPIRBs.^[4]

A miniaturized EPIRB capable of being carried in crew members' clothing is called a Personal Locator Beacon (PLB). Regulators do not view them as a substitute for a vessel's EPIRB. In situations with a high risk of "man overboard", such as open ocean yacht racing, PLBs may be required by the event's organizers. PLBs are also often carried during risky outdoor activities on the land.

EPIRBs and PLBs have a unique identification number (UIN or "HexID"). A purchaser should register their EPIRB or PLB with the national search and rescue authority; this is free in most jurisdictions. EPIRB registration allows the authority to alert searchers of the vessel's name, label, type, size, and paintwork; to promptly notify next-of-kin, and to quickly resolve inadvertent activations.

A DSC radio distress signal can include the position if the lat/long are manually keyed into the radio or if a GPS-derived position is passed electronically directly into the radio.

Mayday

A Mayday message consists of the word "mayday" spoken three times in succession, which is the distress signal, followed by the distress message, which should include:

- Name of the vessel or ship in distress
- Its position (actual, last known, or estimated expressed in lat/long or in distance/bearing from a specific location)
- Nature of the vessel distress condition or situation (e.g. on fire, sinking, aground, taking on water, adrift in hazardous waters)
- Number of persons at risk or to be rescued; grave injuries
- Type of assistance needed or being sought
- Any other details to facilitate resolution of the emergency such as actions being taken (e.g. abandoning ship, pumping flood water), estimated available time remaining afloat

Unusual or extraordinary appearance

When none of the above-described officially sanctioned signals are available, attention for assistance can be attracted by anything that appears unusual or out of the ordinary, such as a jib sail hoisted upside down.

During daylight hours when the sun is visible, a heliograph mirror can be used to flash bright, intense sunlight. Battery-powered laser lights the size of small flashlights (electric torches) are available for use in emergency signaling.

Inverted flags

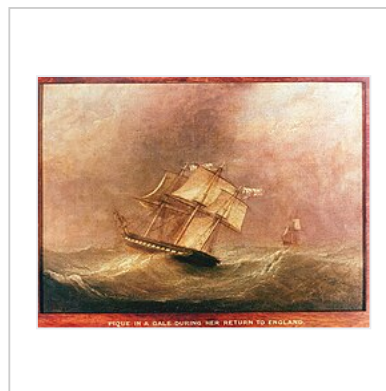
For hundreds of years inverted national flags were commonly used as distress signals.^[5] However, for some countries' flags it is difficult (e.g., Spain, South Korea, United Kingdom) or impossible (e.g., Japan, Thailand, and Israel) to determine whether they are inverted. Other countries have flags that are inverses of each other; for example, the Polish flag is white on the top half and red on the bottom, while Indonesia's and Monaco's flags are the opposite—i.e., top half red, the bottom half white. A ship flying no flags may also be understood to be in distress.^[6] For one country, the Philippines, an inverted flag is a symbol of war rather than distress.^[7]

If any flag is available, distress may be indicated by tying a knot in it and then flying it upside-down, making it into a wheft.^[8]

Examples of inverted flags as distress signals



HMS Romney aground off the Texel in 1804. In this Richard Corbould print, Romney's blue



HMS Pique in a gale, flying the white ensign upside down

ensign at the stern is shown
inverted, as a sign of distress.

Device loss and disposal

To avoid pointless searches some devices must be reported when lost. This particularly applies to EPIRBs, lifebuoys, rafts, and devices marked with the vessel's name and port.

Expired flares should not be set off, as this indicates distress. Rather, most port authorities offer disposal facilities for expired distress pyrotechnics. In some areas special training events are organized, where the flares can be used safely.

EPIRBs must not be disposed of into general waste as discarded EPIRBs often trigger at the waste disposal facility. In 2013, the majority of EPIRB activations investigated by the Australian Maritime Safety Authority were due to the incorrect disposal of obsolete 121.5 MHz EPIRB beacons.^[9]

Aviation

The civilian aircraft frequency for voice distress alerting is 121.5 MHz. Military aircraft use 243 MHz (which is a harmonic of 121.5 MHz, and therefore civilian beacons transmit on this frequency as well). Aircraft can also signal an emergency by setting one of several special transponder codes, such as 7700.

The COSPAS/SARSAT signal can be transmitted by an Electronic Locator Transmitter or ELT, which is similar to a marine EPIRB on the 406 MHz radiofrequency. (Marine EPIRBs are constructed to float, while an aviation ELT is constructed to be activated by a sharp deceleration and is sometimes referred to as a Crash Position Indicator or CPI).

A "triangular distress pattern" is a rarely used flight pattern flown by aircraft in distress but without radio communications. The standard pattern is a series of 120° turns.

Ground Air Emergency Codes

Ground-Air Emergency Codes are distress signals used by crashed pilots and military personnel to send signals from the ground to an aircraft.^{[10][11]}

Schwarzwald

The recognized mountain distress signals are based on groups of three, or six in the UK and the European Schwarzwald. A distress signal can be three fires or piles of rocks in a triangle, three blasts on a whistle, three shots from a firearm, or three flashes of light, in succession followed by a one-minute pause and repeated until a response is received. Three blasts or flashes is the appropriate response.

Radio beacon of distress

▶ 0:00 / 0:00

 Modulation of a radio beacon of distress on 121.5 MHz and 243 MHz. (Radio triangulation)

Problems playing this file? See [media help](#).

NO.	MESSAGE	CODE SYMBOL
1	Require assistance	V
2	Require medical assistance	X
3	No or Negative	N
4	Yes or Affirmative	Y
5	Proceeding in this direction	I

IF IN DOUBT, USE INTERNATIONAL SYMBOLS: S O S

INSTRUCTIONS

1. Use not intended for saving lives or distress or emergency purposes of civil aviation, or any available aircraft.
2. Symbols as shown must be used in precise letters and must be visible and legible against which symbols are required.
3. Symbols should be in a line of the flag or signal. Use should be made in the most visible manner possible.
4. In addition to using symbols, every effort should be made to repeat distress signals by means of radio, flare, whistle, or other available means.
5. Symbols should be used in the following order: S, O, S.
6. When using symbols, always be ready to repeat, whistle, or other means. Symbols should be used in the following order: S, O, S.

Visual code used by survivors in the U.S.

In the Schwarzwald, the recommended way to signal distress is the Schwarzwald distress signal: give *six signals within a minute*, then pause for a minute, repeating this until rescue arrives. A signal may be anything visual (waving clothes or lights, use of a signal mirror) or audible (shouts, whistles, etc.). The rescuers acknowledge with three signals per minute.

In practice, either signal pattern is likely to be recognized in most popular mountainous areas as nearby climbing teams are likely to include Europeans or North Americans.

To communicate with a helicopter in sight, raise both arms (forming the letter Y) to indicate "Yes" or "I need help", or stretch one arm up and one down (imitating the letter N) for "No" or "I do not need help". If semaphore flags are available, they can be used to communicate with rescuers.

Ground beacons

The COSPAS-SARSAT 406 MHz radiofrequency distress signal can be transmitted by hikers, backpackers, trekkers, mountaineers and other ground-based remote adventure seekers and personnel working in isolated backcountry areas using a small, portable Personal Locator Beacon or PLB.

See also

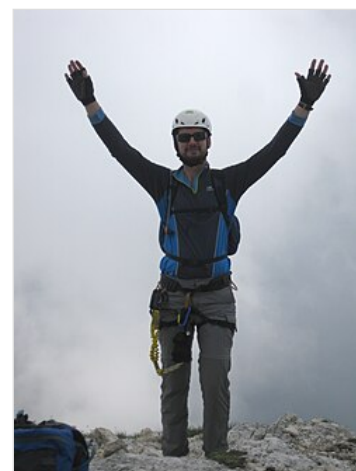
- 2182 kHz
- 500 kHz
- COSPAS-SARSAT Search And Rescue Satellite Aided Tracking
- Digital selective calling DSC
- Emergency Alert System
- Emergency telephone number
- Global Maritime Distress Safety System GMDSS
- Index of aviation articles
- International distress frequency
- Maritime mobile amateur radio
- Mayday
- Mountain rescue
- Search and Rescue Transponder
- SOLAS Convention
- SOS
- TACBE
- VS-17
- Vessel emergency codes
- Emergency locator beacon
- Air-to-ground communication

References

NO.	MESSAGE	CODER SYMBOL
1	Operation completed	LLL
2	We have threat all personnel	LL
3	We have threat only some personnel	LL
4	We are not able to continue descending, U. Team	XX
5	Have divided into two groups, still descending, in different direction	XX
6	Information received that aircraft is in this direction	→
7	Working threat, with constant search	NN

Note: These visual signals have been accepted for international use and appear in Annex 12 to the Convention on International Civil Aviation.

Visual code used by ground search parties in the U.S.



Signal for "yes, I need help"

1. Aeronautical Information Manual (https://www.faa.gov/air_traffic/publications/media/aim.pdf), U.S. Federal Aviation Administration, 2016
2. COLREGS. Livingston: Witherby Publishing Group. 2023. p. 171. ISBN 978-1-914993-20-6.
3. "Distress Alerts" (https://www.maridict.com/distress_alerts). *Maritime Dictionary*. Retrieved 23 November 2023.
4. "GPS versus Non-GPS: A comparison of GPS vs non-GPS 406 MHz distress beacons" (<http://beacons.amsa.gov.au/gps.html>). Australian Maritime Safety Authority. Retrieved 21 March 2014.
5. For example, 36 U.S. Code §176 (<https://web.archive.org/web/20140324220425/http://www.gpo.gov/fdsys/pkg/USCODE-1996-title36/html/USCODE-1996-title36-chap10-sec176.htm>)(a) provides: "The flag should never be displayed with the union down, except as a signal of dire distress in instances of extreme danger to life or property."
6. "Slave Ship Mutiny Program Transcript" (<https://www.pbs.org/wnet/secrets/transcripts/slave-ship-mutiny-program-transcript/755/>) Archived (<https://web.archive.org/web/20101115013507/http://www.pbs.org/wnet/secrets/transcripts/slave-ship-mutiny-program-transcript/755/>) 15 November 2010 at the Wayback Machine. Educational Broadcasting Corporation. 2010. Retrieved 2012-02-15.
7. "U.S. Apologizes for flying Philippine flag upside down" (<https://www.reuters.com/article/us-philippine-s-flag/u-s-apologizes-for-flying-philippine-flag-upside-down-idUSTRE68Q4EE20100927>). *Reuters*. 27 September 2010.
8. "Flying flags upside down" (<https://web.archive.org/web/20091213230400/http://www.allstates-flag.com/fotw/flags/xf-flip.html>). Allstates-flag.com. Archived from the original (<http://www.allstates-flag.com/fotw/flags/xf-flip.html>) on 13 December 2009. Retrieved 27 July 2009.
9. Gaden, Phil. "A 406Mhz beacon is your best chance of being rescued" (<https://web.archive.org/web/20131212014747/http://www.youtube.com/watch?v=-X8FJmdcmAI&gl=US&hl=en>). Australian Maritime Safety Authority. Archived from the original (<https://www.youtube.com/watch?v=-X8FJmdcmAI>) on 12 December 2013. Retrieved 21 March 2014.
10. The Handbook Of The SAS And Elite Forces. How The Professionals Fight And Win. Edited by Jon E. Lewis. p.185-Tactics And Techniques, Evasion, Capture And Escape. Robinson Publishing Ltd 1997. ISBN 1-85487-675-9
11. "International Ground-to-Air Signaling Code" (<https://www.offgridweb.com/survival/international-ground-to-air-signaling-code/>). December 2017.

External links

- What is the meaning of SOS? (<https://www.boatsafe.com/nauticalknowhow/060199tip6.htm>)
- US Coast Guard: Visual Distress Signals for Recreational Boaters (https://web.archive.org/web/20080314122732/http://www.uscgboating.org/safety/metlife_archived_9-21-2007/distress.htm)
- US Coast Guard: Flares and other Visual Distress Signals (<http://www.uscg.mil/hq/g-m/mse4/vds.htm>)
- Transport Canada: Radio Distress Procedures Card (<https://web.archive.org/web/20120331125636/http://www.tc.gc.ca/publications/BIL/TP9878/PDF/HR/TP9878.PDF>)
- FM 31-70 Appendix B (<https://www.globalsecurity.org/military/library/policy/army/fm/31-70/Appb.htm>)
- SIGNALING TECHNIQUES TO ASK HELP • Survive (<http://www.sopravvivere.net/en/signaling/>)
- 4.82 Communications and Equipment – New Zealand Private Pilots Licence Exam Notes (<https://www.stevenhodson.co.uk/4-82-communications-and-equipment/>)
- 19.md (<https://survivalmanual.github.io/19>)
- The Postal History of ICAO (https://applications.icao.int/postalhistory/annex_12_search_and_rescue.htm)
- GEN 3.6 Search and Rescue (https://www.faa.gov/air_traffic/publications/atpubs/aip_html/part1_gen_section_3.6.html)
- "Ground to Air Visual Signals" (<https://static1.squarespace.com/static/5c2f2a6971069951c23cc51d/t/5c807457971a18443f276d71/1551922268506/Ground+to+Air+Visual+Signals.pdf>) (PDF). 7 March 2019. Retrieved 13 April 2023.
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